

An elastic network model whose parameters are rest lengths

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Abstract

In a standard elastic network model the spring constants are set by the equilibrium distances of a given structure, which makes the contact map a property of that structure. This is harmless for a single native state but awkward for mutation, where the object that changes is the sequence, not the coordinates. We formulate an ENM in which the rest lengths l_{ij} are the only parameters and the spring constants follow them, $k_{ij} = k(l_{ij})$. The contact map then becomes a property of the parameters, contacts form and break through mutation rather than through relaxation, and the mutant structure is defined as the minimiser of the mutant potential rather than by a linear response. We derive the resulting theory for a single mutation from an arbitrary reference: the exact decomposition of the mutation energy into three terms, two of which vanish when the reference is the relaxed wild type and neither of which vanishes in general; and the Hessian, whose transverse term is the sole channel through which frustration reaches the dynamics. The two vanishing terms are exactly what a treatment calibrated at the wild type discards.

1 Introduction

An elastic network model (ENM) represents a protein as beads joined by harmonic springs, and reproduces the slow collective motions of the native state remarkably well for so crude a description. In the standard construction one is handed an experimental structure $\mathbf{r}^e$, declares two residues in contact when their separation falls below a cutoff, and assigns each contact a spring whose rest length is the distance observed in that structure. The equilibrium conformation is then the input structure by construction, and the model is strain-free: every spring sits at its rest length simultaneously.

This construction ties the model to a structure. That is exactly right when the question is the dynamics of one native state, and it becomes awkward as soon as the question is mutation. A mutation changes the sequence; the structure is a *consequence*. But in the standard formulation the contact map — which springs exist at all — is read off the coordinates, so there is no way to state what a mutation does to the model without first knowing what it does to the structure. The usual resort is to hold the contact map fixed and perturb within it, which forbids contacts from forming or breaking, or to rebuild the model from a new structure, which presupposes the answer.

There is a second difficulty, less often noticed. Once mutations accumulate, the reference state is no longer the relaxed wild type. It is a structure that has been pushed away from every spring's preference and settled at a compromise: a *frustrated* state, in which no conformation relaxes all springs at once and the residual strain is nonzero at the minimum. Expressions derived at the wild type routinely assume a relaxed reference — often silently, through a term that happens to vanish there. Applied to a strained reference they are no longer exact, and the error is not a small correction of fixed sign but a term proportional to the accumulated strain.

This note develops an ENM designed around both points. The move is small: make the rest lengths the parameters, and let the spring constants be a function of the rest lengths,

$$k_{ij} = k(l_{ij}) \quad \text{rather than} \quad k_{ij} = k(d_{ij}^e), \quad (1)$$

where d_{ij}^e is the distance in the equilibrium structure. At the wild type the two agree, because the standard fit sets $l_{ij} = d_{ij}^e$; they part company for a mutant, and the consequence is the point of the model:

the contact map becomes a property of the parameters, not of the structure.

Contacts are made and broken by *mutation*, when a rest length crosses the cutoff — not by relaxation. A mutation is then a well-defined operation on the model alone, and the mutant structure follows from it rather than being required in advance.

We treat a single mutation from an arbitrary reference, which may itself be strained. Section 2 sets out the potential and the state, Section 3 defines a mutation and the mutant structure, Section 4 derives the energy differences and the exact decomposition that is the main formal result, and Section 5 derives the Hessian and isolates the transverse term. Throughout, $k(l)$ is an arbitrary non-negative function; the hard cutoff of the anisotropic network model (ANM) appears in Section 6 as a special case, and is the one choice that makes the energy discontinuous in the parameters.

Notation

N	number of beads (residues), one per site
$\mathbf{r}$	a conformation, the $3N$ coordinates $(\mathbf{r}_1, \dots, \mathbf{r}_N)$
$d_{ij}(\mathbf{r})$	$= \ \mathbf{r}_i - \mathbf{r}_j\ $, the distance between beads i and j
l_{ij}	rest length of the spring on pair ij ; the model's parameters
$k(l)$	spring constant as a function of rest length, $k \geq 0$
$d_{\max}$	contact cutoff, the scale at which $k(l)$ falls to zero
$\mathbf{r}^e$	an equilibrium (relaxed) conformation, a minimiser of V
$V(\mathbf{r}; l)$	the potential, eq. (2)
$\hat{\mathbf{e}}_{ij}$	$= (\mathbf{r}_i - \mathbf{r}_j)/d_{ij}$, the unit vector along pair ij
$\mathbf{I}$	the 3×3 identity

Subscripts *ref* and *mut* label the reference and the mutant; $\mathbf{r}^e$ with such a subscript is that model's own relaxed structure. We write l for the whole parameter set $\{l_{ij}\}$ when no ambiguity arises.

2 The model

2.1 The potential

The potential energy of a conformation $\mathbf{r}$ under parameters l is

$$V(\mathbf{r}; l) = \frac{1}{2} \sum_{i < j} k(l_{ij}) (d_{ij}(\mathbf{r}) - l_{ij})^2. \quad (2)$$

Each pair contributes a harmonic term penalising departure of its current distance from its rest length, with a stiffness fixed by that same rest length. We take the zero of energy at the fully relaxed configuration, so that V evaluated at any conformation is exactly the strain energy stored in the springs.

The sum runs over all $\frac{1}{2}N(N-1)$ distinct pairs. Pairs with $k(l_{ij}) = 0$ contribute nothing and are *inactive*; those with $k(l_{ij}) > 0$ are *active*, and the set of active pairs is the contact map. Because k is evaluated at l_{ij} and not at $d_{ij}(\mathbf{r})$, membership of the contact map is determined by the parameters alone, independently of the conformation at which V is being evaluated. This is the defining property of the model and every result below traces back to it.

2.2 States

A state of the model is a complete assignment of rest lengths, one for every pair of sites — not merely for the active ones. The wild-type state is obtained from an experimental structure $\mathbf{r}_{\text{wt}}^e$ by

$$l_{ij}^{\text{wt}} = d_{ij}(\mathbf{r}_{\text{wt}}^e), \quad (3)$$

which is the standard ENM fit, and reproduces the standard model exactly: every spring sits at its rest length at $\mathbf{r}_{\text{wt}}^e$, so the wild-type structure is a minimum of V with zero strain energy, and $k(l_{ij}^{\text{wt}}) = k(d_{ij}^e)$ recovers the usual assignment.

Carrying rest lengths for inactive pairs is forced by the requirement that contacts be able to form. By eq. (3) every active wild-type pair has a rest length inside the cutoff; if inactive pairs carried no parameter, no pair outside the cutoff could ever be brought inside one, and mutation could only remove contacts. Assigning every pair a rest length makes formation and breaking symmetric operations. The enlargement is one of parameters only: inactive pairs have $k = 0$ and contribute neither energy nor curvature, so the Hessian retains its $3N \times 3N$ form and the active pairs alone determine the dynamics.

2.3 Strain and frustration

For a state l with equilibrium structure $\mathbf{r}^e$, define the residual of a pair as the amount by which its spring is stretched or compressed at equilibrium,

$$r_{ij} \equiv d_{ij}(\mathbf{r}^e) - l_{ij}, \quad (4)$$

and the strain energy as $E_{\text{strain}} = V(\mathbf{r}^e; l)$, which by eq. (2) is $\frac{1}{2} \sum_{i < j} k(l_{ij}) r_{ij}^2$.

A state is **frustrated** when no conformation relaxes every active spring at once, equivalently when $E_{\text{strain}} > 0$ at the minimum. The wild-type state is unfrustrated by construction: eq. (3) makes every residual vanish at $\mathbf{r}_{\text{wt}}^e$. A mutant is frustrated in general, because there is no reason a perturbed set of rest lengths should be simultaneously realisable by any embedding of N points in three dimensions. Frustration is thus the generic condition of the model and the unfrustrated wild type is the special case — the reverse of the usual situation, and the reason the terms identified in Section 4 cannot be dismissed.

3 Mutation

A mutation at site s perturbs the rest lengths of all pairs involving that site,

$$l_{ij}^{\text{mut}} = l_{ij}^{\text{ref}} + \delta l_{ij}, \quad \delta l_{ij} \sim \mathcal{N}(0, \sigma^2) \text{ for } i = s \text{ or } j = s, \quad (5)$$

and leaves all other rest lengths unchanged. The parameter σ sets the mutation size. The spring constants are not free: they follow the rest lengths through eq. (1), so the mutation induces

$$\delta k_{ij} = k(l_{ij}^{\text{ref}} + \delta l_{ij}) - k(l_{ij}^{\text{ref}}), \quad (6)$$

which is nonzero precisely for those pairs whose rest length has moved into a region of different stiffness — in particular, across the cutoff. Contacts form and break here, in the parameters, and nowhere else.

The mutant structure is defined as a minimiser of the mutant potential,

$$\mathbf{r}_{\text{mut}}^e = \arg \min_{\mathbf{r}} V(\mathbf{r}; l^{\text{mut}}), \quad (7)$$

and the mutant's energy is $V(\mathbf{r}_{\text{mut}}^e; l^{\text{mut}})$, read at that minimum. Two features of this definition are worth making explicit, because they are what distinguish the model from the usual perturbative treatments.

First, **no expansion conditions the structure**. Eq. (7) is a statement about the exact potential, not about a quadratic approximation to it; the mutant structure is not $\mathbf{r}_{\text{ref}}^e$ displaced by a linear response $\mathbf{C}_{\text{ref}}\mathbf{f}$, and it is not obtained by truncating anything. Normal modes, when computed, describe a state that has already been determined; they never determine it.

Second, **the contact map may differ between reference and mutant**, by eq. (6). The comparison in Section 4 is therefore between two genuinely different Hamiltonians — not merely between two parameter values of one Hamiltonian — and the bookkeeping must accommodate springs that exist on one side and not the other.

4 Energy differences

4.1 Three differences

Comparing a mutant to its reference involves two potentials and two structures, so a difference in energy is ambiguous until both are named. Three combinations are meaningful. The *stress* energy is the cost of changing the Hamiltonian while holding the structure fixed at the reference:

$$\Delta V_{\text{stress}} = V(\mathbf{r}_{\text{ref}}^e; l^{\text{mut}}) - V(\mathbf{r}_{\text{ref}}^e; l^{\text{ref}}). \quad (8)$$

The *relaxation* energy is the energy released when the mutant Hamiltonian is allowed to find its own minimum:

$$\Delta V_{\text{relax}} = V(\mathbf{r}_{\text{mut}}^e; l^{\text{mut}}) - V(\mathbf{r}_{\text{ref}}^e; l^{\text{mut}}) \leq 0. \quad (9)$$

It is non-positive because $\mathbf{r}_{\text{mut}}^e$ minimises $V(\cdot; l^{\text{mut}})$ by eq. (7). Their sum is the difference of minima,

$$\Delta V_{\text{min}} = V(\mathbf{r}_{\text{mut}}^e; l^{\text{mut}}) - V(\mathbf{r}_{\text{ref}}^e; l^{\text{ref}}) = \Delta V_{\text{stress}} + \Delta V_{\text{relax}}, \quad (10)$$

which compares each state at its own equilibrium and is the quantity a selective criterion would act on. In words: two Hamiltonians at one structure; one Hamiltonian at two structures; and the difference of minima.

4.2 The exact decomposition

The stress energy can be written out exactly. Substituting $l_{ij}^{\text{mut}} = l_{ij}^{\text{ref}} + \delta l_{ij}$ and $k_{ij}^{\text{mut}} = k_{ij}^{\text{ref}} + \delta k_{ij}$ into eq. (8), with r_{ij} the reference's own residual from eq. (4), gives

$$\Delta V_{\text{stress}} = \underbrace{\frac{1}{2} \sum_{i < j} k_{ij}^{\text{ref}} \delta l_{ij}^2}_{(a)} - \underbrace{\sum_{i < j} k_{ij}^{\text{ref}} r_{ij} \delta l_{ij}}_{(b)} + \underbrace{\frac{1}{2} \sum_{i < j} \delta k_{ij} (r_{ij} - \delta l_{ij})^2}_{(c)}. \quad (11)$$

This is an identity, not an expansion: no term has been truncated, and it holds for arbitrary $k(l)$, arbitrary mutation size, and an arbitrarily strained reference. The three terms have distinct characters.

(a) The direct cost. Always present and always positive, this is the work done stretching the springs that already exist, and it is the only term that survives when the reference is relaxed and the contact map is unchanged. It is quadratic in the mutation size and independent of the reference's strain.

(b) The cross term. This term couples the mutation to the strain the reference already carries. It vanishes if and only if the reference is relaxed, $r_{ij} = 0$ — in particular it vanishes at the wild type, by eq. (3). It is the only term of the three that can be negative: a mutation that happens to move rest lengths toward the distances the structure already exhibits releases energy. Its magnitude is proportional to the reference's accumulated strain, so it is negligible near the wild type and grows as the reference becomes more frustrated. A formula calibrated at the wild type therefore appears correct, and degrades in exactly the regime where mutations have accumulated.

(c) The contact term. This term exists only because k follows l . It vanishes if and only if $\delta k_{ij} = 0$ for all pairs, which is the assumption made by models that perturb within a fixed contact map — the linearly forced ENM (LFENM) among them. Here a pair whose stiffness changes contributes its whole stored energy, and if $k(l)$ is a step then that contribution arrives discontinuously (Section 6). Note that (c) is governed by $r_{ij} - \delta l_{ij}$, the residual the pair *would* have under the mutant rest length, so a contact forming far from its new rest length is expensive.

Terms (b) and (c) are precisely what is discarded by a treatment calibrated at the relaxed wild type with a frozen contact map. Neither is a higher-order correction that could be recovered by expanding further; both are exact contributions that happen to vanish in that special case.

4.3 Differences, not energies

One consequence deserves emphasis because it is a standing source of error. ΔV_{stress} is a *difference*, and it is not interchangeable with the absolute energy $V(\mathbf{r}_{\text{ref}}^e; l^{\text{mut}})$ of the mutant Hamiltonian at the reference structure. The two differ by $V(\mathbf{r}_{\text{ref}}^e; l^{\text{ref}})$, the reference's own strain energy, which is zero at the wild type and grows as the reference becomes frustrated. At the wild type the distinction is invisible; away from it, it is the whole of the reference's accumulated strain.

5 Dynamics

5.1 The Hessian

The motions of a state are governed by the second derivatives of V at its equilibrium structure. Differentiating eq. (2) twice with respect to the coordinates gives, for the off-diagonal 3×3 block coupling two active sites i and j ,

$$\mathbf{K}_{ij} = -k(l_{ij}) \left[\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^T + g_{ij} (\hat{\mathbf{e}}_{ij} \hat{\mathbf{e}}_{ij}^T - \mathbf{I}) \right], \quad g_{ij} = \frac{l_{ij}}{d_{ij}(\mathbf{r})} - 1, \quad (12)$$

with the diagonal blocks fixed by translational invariance, $\mathbf{K}_{ii} = -\sum_{j \neq i} \mathbf{K}_{ij}$. Inactive pairs have $k(l_{ij}) = 0$ and drop out, so the matrix is $3N \times 3N$ regardless of how many pairs carry parameters. It has six zero eigenvalues, corresponding to the three translations and three rotations under which V is invariant.

5.2 The transverse term

Eq. (12) contains two pieces. The first, $\hat{\mathbf{e}}\hat{\mathbf{e}}^\top$, is the longitudinal projector: it resists stretching along the line joining the two sites and is the whole of the standard ENM Hessian. The second, weighted by g_{ij} , acts in the plane *perpendicular* to that line, and is the **transverse term**. Its coefficient

$$g_{ij} = \frac{l_{ij}}{d_{ij}(\mathbf{r})} - 1 \quad (13)$$

measures the fractional mismatch between the spring's rest length and its actual extension. A stretched spring ($d_{ij} > l_{ij}$) has $g_{ij} < 0$ and resists transverse displacement, like a taut string; a compressed one has $g_{ij} > 0$ and assists it.

The transverse term vanishes identically when every spring sits at its rest length. This gives the model's sharpest structural statement:

frustration reaches the dynamics only through g_{ij} .

A state's residuals enter the energy through the strain E_{strain} and enter the motions through g_{ij} , and through nothing else. Standard ENMs omit the transverse term, and are entitled to: their fit, eq. (3), forces $l_{ij} = d_{ij}(\mathbf{r}^e)$, hence $g_{ij} = 0$ at the structure where the Hessian is evaluated. The omission is exact for an unfrustrated state and is an approximation of uncontrolled sign for any other — including every mutant of the present model.

It is worth distinguishing the transverse term from the cross term (b) of Section 4.2, since both vanish at a relaxed reference and both are proportional to residuals. They are different objects: (b) is a contribution to an energy difference, linear in the mutation; g_{ij} is a contribution to the curvature at a single state, independent of any mutation. A state can have no mutation applied to it and still have $g_{ij} \neq 0$.

6 The step-function case

The theory above holds for arbitrary $k(l)$. The ANM corresponds to the hard cutoff

$$k(l) = k_0 \mathbb{1}[l \leq d_{\max}], \quad (14)$$

in which a pair is either a full-strength spring or absent. This choice is the common one and it has a specific consequence: V is *discontinuous* in the parameters. A rest length crossing $d_{\max}$ adds or removes a spring outright, and by term (c) of eq. (11) the energy jumps by $\frac{1}{2}k_0(r_{ij} - \delta l_{ij})^2$, an amount that does not go to zero as the crossing becomes marginal. Two states with almost identical parameters can have appreciably different energies.

Nothing in Sections 2–5 requires this. A smooth $k(l)$ — a sigmoid of width w centred on $d_{\max}$, say — makes V continuous and differentiable in the parameters, makes δk_{ij} small whenever δl_{ij} is small, and reduces term (c) to a smooth contribution rather than a jump. The price is that the contact map ceases to be a crisp binary object: every pair carries some stiffness, and "the contact map" becomes a thresholded description of a continuum rather than a property of the state. The step and the sigmoid are thus two ends of a modelling choice about how sharply stiffness should fall with rest length, and the theory is written to accommodate either.

7 Discussion

The model is a small departure from the standard ENM — one argument changed, from $k(d_{ij}^e)$ to $k(l_{ij})$ — and the consequences follow from that single change. Making the rest lengths the parameters makes the contact map a property of the parameters; that in turn makes mutation a well-defined operation on the model that can create and destroy contacts, and makes the mutant structure a derived quantity rather than an input.

The formal content is eq. (11) and eq. (12). Both exhibit the same pattern: a term that is present in general, plus terms that vanish precisely when the reference is the relaxed wild type with a fixed contact map. In the energy these are the cross term (b), proportional to accumulated strain, and the contact term (c), proportional to changes in stiffness. In the dynamics it is the transverse term g_{ij} . A treatment calibrated at the wild type sets all three to zero and is exactly right in doing so — at the wild type. The model’s purpose is to describe states reached by mutation, where none of the three is zero, and where the discarded terms grow with the very quantity that accumulates.

Two limitations of scope should be stated. We have treated a single mutation from a given reference; the reference may be arbitrarily strained, so the results apply at any point of a substitution process, but the process itself — how mutations are proposed, accepted, and accumulated — is not developed here, and neither are the statistics of strain along such a process. And the theory is formal throughout: it establishes what the terms are and when they vanish, not how large they are in any particular protein, which is an empirical question the decomposition is designed to make answerable.